

Program Development Process

Problem Specification

What exactly is the problem?
How will the program be used?
How will the program behave?

Problem Decomposition

What objects will be used and how
will they interact with each other?

*The problem is divided into
objects. For each object we
design a class.*

Class Design

What role or roles will the object perform?
What variables (attributes) will it need?
What methods (behaviors) will it use?
What interface will it present?
What information will it hide?

*The object's role
decomposes into
tasks. Each task
can be assigned to
a method.*

Data Design

What types of instance variables
are needed?
Should they be public or private?

Method Design

What task will the method perform?
What information will it need?
What algorithm will it use?
What result will it produce?

Algorithm Design

What information is needed?
What control structures are needed?

*Method design involves
designing an algorithm.*

Coding into Java

Stepwise refinement
Fixing syntax errors

*Errors may require
recoding or
redesigning.*

Testing, Debugging, Revising

Designing test data and test cases
Fixing semantic errors

